1. Provide a definition or explanation for the term "decomposition". (5 points)

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5

Decomposition is breaking a task or program up into parts, each of which is simpler than the whole. Decomposing a program may involve giving a group of statements an identity or name, so it can easily be called again. Decomposition helps eliminate redundency and makes programs clearer or easier to understand. For example theses statements

```
System.out.println("Hello world!");
System.out.println("This is my first program.");
```

might be both called using the static method createIntro().

excellent

1. Provide a definition or explanation for the term "decomposition". (5 points)

5

De composition is the process of breaking something down (a program) into smaller and more comprehendable parts or methods. This process is used a lot in coding to make writing long complex programs more simple. For example;

```
public static void main() {
    drawRectangle();   ← condensed method
}

public static void drawRectangle() {
    System.out.println("--------")
    System.out.println("|      |")
    System.out.println("|      |")
    System.out.println("|------|")
    System.out.println
}
```

OK, EXAMPLE BUT YOU'VE MISSED THE OPPORTUNITY HERE TO HIGHLIGHT THESE 2 PARTS

2. Consider the following complete and working Java source program.

```
 1 public class Quiz01v1  {
 2
 3    public static void main( String [] args ) {
 4       System.out.println("PMJ's Quiz 01 Fall 2019 ...");
 5       prefix();  bar();   suffix();  System.out.println();
 6       prefix();  logo();  suffix();  System.out.println();
 7       prefix();  bar();   suffix();  System.out.println();
 8       next();    next();  System.out.println();
 9       System.out.println();
10       next();
11    }
12
13    public static void logo() {
14       prefix();
15       System.out.print(" UofS ");
16       suffix();
17    }
18
19    public static void prefix() {
20       System.out.print("<<");
21    }
22
23    public static void bar() {
24       System.out.print("==========");
25    }
26
27    public static void suffix() {
28       System.out.print(">>");
29    }
30
31    public static void next() {
32       System.out.print("!");
33    }
34 }
```

In the space provided below, provide the complete and precise output that results from the execution of the above program. You are to place one character per block, and give the exact output as clearly as you can. (5 points)

```
PMJ's Quiz 01 Fall 2019 ...
<<==========>>
<<<< UofS >>>>
<<==========>>
!!

!
```

PMJ's Quiz 01 Fall 2019 ...

<<==========>>

<<<< UofS >>>>

<<==========>>

!!

!